Radha Kumaran

Research Interests: Mixed Reality, Multimodal Interaction, Intelligent User Interfaces

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in *radha-kumaran*

Academic Positions

2026-present **Saarland University** Saarbrücken, Germany
Postdoctoral Researcher
Supervisor: Prof. Dr. Jürgen Steimle

Education

2019-2025 **University of California, Santa Barbara** Santa Barbara, CA, USA
PhD in Computer Science
Advisor: Prof. Tobias Höllerer
Dissertation: Supporting search in mobile augmented reality.

2015-2019 **R.V. College of Engineering** Bangalore, India
B.E, Computer Science and Engineering

Peer-Reviewed Conference and Journal Publications

2026 Arthur Caetano, **Radha Kumaran**, Kelvin Jou, Tobias Höllerer and Misha Sra. 2026. XARP: A Human-First and Agent-Ready Extended Reality Toolkit in Python. To appear in *Proceedings of the ACM on Human-Computer Interaction EICS*, 32 pages. *[pdf] [link]*

2026 Tom Bullock, Emily Machniak, You-Jin Kim, **Radha Kumaran**, Justin Kasowski, Apurv Varshney, Julia Ram, Melissa M Hernandez, Stina Johansson, Neil M Dundon, Tobias Höllerer and Barry Giesbrecht. 2026. SABER: Spatial Attention, Brain, Extended Reality. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 10 pages. *[pdf] [link]*

2025 Kangyou Yu, Yunhao Luo, **Radha Kumaran**, Shane Dirksen, Misha Sra, and Tobias Höllerer. 2025. The Cost of Virtuality Switching: Searching for Physical and Virtual Targets in Optical-See-Through Augmented Reality. In *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 13 pages. *[pdf] [link]*

2025 Tom Bullock, Emily Machniak, Joyce Passananti, Kangyou Yu, **Radha Kumaran**, You-Jin Kim, Jordan Garrett, Madhav Viswesvaran, Julia Ram, Tobias Höllerer, and Barry Giesbrecht. 2025. Electrophysiological evidence for differential semantic processing of words and objects presented in augmented reality. In *Computers in Human Behavior Reports, Volume 18*, 12 pages. *[pdf] [link]*

2025 You-Jin Kim, **Radha Kumaran**, Jingjing Luo, Tom Bullock, Barry Giesbrecht, and Tobias Höllerer. 2025. On the Go with AR: Attention to Virtual and Physical Targets while Varying Augmentation Density. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 16 pages. *[pdf] [link]*

2024 Chengyuan Xu*, **Radha Kumaran***, Noah Stier, Kangyou Yu, and Tobias Höllerer. 2024. Multimodal 3D Fusion and In-Situ Learning for Spatially Aware AI. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 10 pages. *[pdf] [link]*

2023 Jiarui Zhu, **Radha Kumaran**, Chengyuan Xu, and Tobias Höllerer. 2023. Free-form Conversation with Human and Symbolic Avatars in Mixed Reality. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 10 pages. *[pdf] [link]*

2023 **Radha Kumaran***, You-Jin Kim*, Anne Milner, Tom Bullock, Barry Giesbrecht, and Tobias Höllerer. 2023. The Impact of Navigation Aids on Search Performance and Object Recall in Wide-Area Augmented Reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 17 pages. *[pdf] [link]*

2022 You-Jin Kim*, **Radha Kumaran***, Ehsan Sayyad, Anne Milner, Tom Bullock, Barry Giesbrecht, and Tobias Höllerer. 2022. Investigating Search Among Physical and Virtual Objects Under Different Lighting Conditions. In *IEEE Transactions on Visualization and Computer Graphics 28, no. 11 (2022)*: 3788-3798, 11 pages. *[pdf] [link]*

2020 Yi Ding, **Radha Kumaran**, Tianjiao Yang, and Tobias Höllerer. 2020. Predicting Video Affect via Induced Affection in the Wild. In *Proceedings of the 2020 International Conference on Multimodal Interaction (ICMI '20)*, 10 pages.*[pdf] [link]*

2017 Ramya Ramanathan, **Radha Kumaran**, Ram Rohan Ramanathan, Rajat Gupta, and Vishalakshi Prabhu. 2017. An Intelligent Music Player Based on Emotion Recognition. In *2017 2nd International Conference on Computational Systems and Information Technology for Sustainable Solution (CSITSS)*, 5 pages. *[pdf] [link]*

* indicates equal contribution.

Juried Non-Archival Abstracts, Posters and Workshop Papers

2026 **Radha Kumaran**, You-Jin Kim, Shane Dirksen, Tom Bullock, Barry Giesbrecht, and Tobias Höllerer. 2026. Challenges and Opportunities for Head-Worn AR Use in Wide-Area Environments. In *CHI 2026 Workshop on Next Steps for Augmented Reality On-the-Move: Challenges & Opportunities*, 2 pages. *[link]*

2025 Shane Dirksen, **Radha Kumaran**, You-Jin Kim, Yilin Wang, and Tobias Höllerer. 2025. Modeling Object Attention in Mobile AR for Intrinsic Cognitive Security. In *1st Workshop on Enhancing Security, Privacy, and Trust in Extended Reality (XR) Systems*, 6 pages. *[pdf][link]*

2025 Tom Bullock, Emily Machniak, Neil Dundon, Justin Kasowski, You-Jin Kim, **Radha Kumaran**, Julia Ram, Melissa Hernandez, Stina Johansson, Tobias Höllerer and Barry Giesbrecht. 2025. Reconstructing moving object locations in an immersive 3D virtual environment from EEG oscillatory activity. In *Journal of Vision 25 (9)*, 1 page. *[pdf][link]*

2025 **Radha Kumaran**, You-Jin Kim, Emily Machniak, Shane Dirksen, Junhyung Yoon, Tom Bullock, Barry Giesbrecht, and Tobias Höllerer. 2025. Scene Awareness While Using Multiple Navigation Aids in AR Search. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2 pages. *[pdf][link]*

2023 **Radha Kumaran***, Viral Niraj Doshi*, Sherry X. Chen, Avinash Ajit Nargund, Tobias Höllerer, and Misha Sra. 2023. EChat: An Emotion-Aware Adaptive UI for a Messaging App. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, 3 pages. *[pdf] [link]*

* indicates equal contribution.

Preprints

2025 Arthur Caetano, Kavya Verma, Atieh Taheri, **Radha Kumaran**, Zichen Chen, Jiaao Chen, Tobias Höllerer, and Misha Sra. 2025. Agentic Workflows for Conversational Human-AI Interaction Design. arXiv preprint arXiv:2501.18002, 14 pages. [link]

Teaching

2024 **Guest Lecturer for CS 185: Human-Computer Interaction**
UC Santa Barbara, Dept. of Computer Science
Instructor: Misha Sra (W24)
- Gave a lecture on augmented reality interaces to support search tasks

2020, 2023 **TA for CS 8: Introduction to Computer Science**
UC Santa Barbara, Dept. of Computer Science
Instructors: Yekaterina Kharitonova (W20, F20) and Diba Mirza (S20, M20, M23)
- Helped to create and/or revise course material, including quizzes, homeworks and final projects
- Prepared material for and led discussion sections every week
- Graded quizzes, midterms and finals
- Held office hours and/or open labs every week, helping students with understanding problems, forming solutions and debugging as needed

2019 **TA for CS 176A: Introduction to Computer Communication Networks**
UC Santa Barbara, Dept. of Computer Science
Instructor: Elizabeth Belding (F19)
- Graded quizzes, midterms and finals
- Held office hours and open labs every week, helping students with understanding problems, forming solutions and debugging as needed

Mentorship

Masters Students

2024-2025 Sangita Kunapuli, BS/MS at UCSB

2023 Jiarui Zhu (now at Waymo), Satyam Awasthi (now at Yahoo), Vivian Ross (now at Amazon)

Undergraduate Students

2024-2025 Junhyung Yoon, Sophomore at UCSB

2023-2024 Ajay Liu, Towela Phiri, Frank Zhong, Qi Wu (UCSB CS Early Research Scholars)

2022 Viral Niraj Doshi (now at Generac)

High School Students

2022 Anika Sikka (now UC Berkeley Class of '27), Woyu Wang (now UCLA Class of '27)

Invited Talks

2025 **Supporting Search in Mobile Augmented Reality**, Intelligent Inclusive Interaction Design (I3D) Lab, Indian Institute of Science

2025 **Augmented Search: Insights from Three Mobile Augmented Reality Studies**, UCSB/University of Toronto Human and Computer Vision Workshop

Fellowships and Awards

2024 **Outstanding Mentor of the Year**, UCSB Computer Science Department

2023 **Inclusion, Diversity, Equity, and Accessibility (IDEA) Scholarship**, IEEE ISMAR

2023 **Gary Marsden Travel Award**, ACM SIGCHI

2023 **Doctoral Student Travel Award**, University of California, Santa Barbara

2019 **Academic Excellence Fellowship**, University of California, Santa Barbara

2018 **Best Hackathon Project Nationwide**, IBM India

Service and Leadership

2025, 2026 **Organizer**, Future Faculty Forum at IEEE VR

2024, 2025 **Organizer**, Future Faculty Forum at IEEE ISMAR

2024-2025 **Graduate Diversity Representative**, UCSB CS Department

2022-2024 **Vice President, Research**
UCSB Women in Computer Science (WiCS)

2021-2022 **Director, Research**
UCSB WiCS

Program Committee Member

2026 CHI, ISMAR, VRST

2025 ISMAR

External Reviewing

2026 DIS, IEEE VR

2025 CHI*, DIS, IEEE VR, TVCG

2024 CHI, ISMAR, SUI, ISS, VRST, TVCG

2023 CHI LBW, IEEE VR, ISMAR, VRST

* indicates special recognition for useful reviews

Volunteering

2024 **UCSB Hacks**; judge

2023, 2024 **UCSB CS PhD Application Support Program**; volunteer reviewer

2023, 2024 **ISMAR**; student volunteer

2023 **WiCS Outreach**; guest lecturer at local high school